

Decent Coaching Classes

VIII SB

Mind Your Math - II

Answer Key: Test - Chapters 4 and 6

Q.1 (A) Choose the correct alternative

- (1) (B) orthocentre
- (2) (C) 2 : 1
- (3) (A) $(x + a)(x + b)$
- (4) (A) $(a - b)(a^2 + ab + b^2)$

Q.1 (B) Solve the following (any four)

- (1) The orthocentre of a right-angled triangle lies at the vertex of the right angle.
- (2) The orthocentre of an obtuse-angled triangle lies in the exterior of the triangle.
- (3) $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- (4) $y^2 - 10y + 9 = (y - 9)(y - 1)$
- (5) $l(AG) : l(GD) = 2 : 1 \Rightarrow l(GD) = l(AG) \div 2 = 8 \div 2 = 4 \text{ cm}$

Q.2 Activities (any two)

- (1) $2x^2 + 5x - 18 = 2x^2 + 9x - 4x - 18 = x(2x + 9) - 2(2x + 9) = (2x + 9)(x - 2)$
[blanks: 9 and -4; then 9x, 4x; then $(2x + 9)$, $(2x + 9)$; final factor $(2x + 9)$]
- (2) $(2p + 3m + 4n)^2 = 4p^2 + 9m^2 + 16n^2 + 12pm + 24mn + 16pn$
[blanks: $4p^2$, $16n^2$, $24mn$, $16pn$]
- (3) $AG : GD = 2 : 1$; $BE = 12 \text{ cm} \Rightarrow BG = (2/3) \times 12 = 8 \text{ cm}$ (since $BG : GE = 2 : 1$)

Q.3 Solve the following (any four)

- (1) $x^2 - 10x + 9 = (x - 9)(x - 1)$
- (2) $p^2 - 2p - 35 = (p - 7)(p + 5)$
- (3) $(3x + 2y)^3 - (3x - 2y)^3 = 6(3x)^2(2y) + 2(2y)^3 = 108x^2y + 16y^3$
- (4) $3x^2 + 14x + 15 = 3x^2 + 9x + 5x + 15 = 3x(x + 3) + 5(x + 3) = (x + 3)(3x + 5)$
- (5) Since $\triangle PQR$ is equilateral, the altitude and median from any vertex coincide (by symmetry, the foot of the perpendicular from P is also the midpoint of QR). Hence S and T are the same point.
- (6) $8p^3 + 125q^3 = (2p)^3 + (5q)^3 = (2p + 5q)(4p^2 - 10pq + 25q^2)$

Q.4 Solve the following (any two)

- (1) (i) $AG : GD = 2 : 1$, $AD = 15 \text{ cm} \Rightarrow AG = 10 \text{ cm}$, $GD = 5 \text{ cm}$
(ii) $BG : GE = 2 : 1$, $GE = 4 \text{ cm} \Rightarrow BG = 8 \text{ cm}$, $BE = BG + GE = 12 \text{ cm}$
(iii) The centroid divides each median in the ratio 2 : 1 from the vertex. Yes, the centroid always lies in the interior of the triangle, for acute, right, and obtuse triangles alike.
- (2) (i) $20x^2 - 26x + 8 = 2(10x^2 - 13x + 4) = 2(10x^2 - 8x - 5x + 4) = 2[2x(5x - 4) - 1(5x - 4)] = 2(5x - 4)(2x - 1)$
(ii) $125y^3 - 1 = (5y)^3 - 1^3 = (5y - 1)(25y^2 + 5y + 1)$
(iii) $(a + b)^3 - a^3 - b^3 = (a^3 + 3a^2b + 3ab^2 + b^3) - a^3 - b^3 = 3a^2b + 3ab^2 = 3ab(a + b)$
- (3) $p^2 - 16q^2 = (p - 4q)(p + 4q)$; $p^3 - 64q^3 = (p - 4q)(p^2 + 4pq + 16q^2)$
Simplified expression = $(p + 4q) / (p^2 + 4pq + 16q^2)$
- (4) Since $AB = AC$, $\triangle ABC$ is isosceles about vertex A. By the symmetry of an isosceles triangle, the perpendicular from A to BC also bisects BC and bisects $\angle A$. So the altitude, median and angle-bisector from A are all the same segment. This segment is also the perpendicular bisector of BC. Hence the circumcentre, incentre, centroid and orthocentre of $\triangle ABC$ all lie on this one segment — that is, they are collinear.